JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to KRISHNA AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

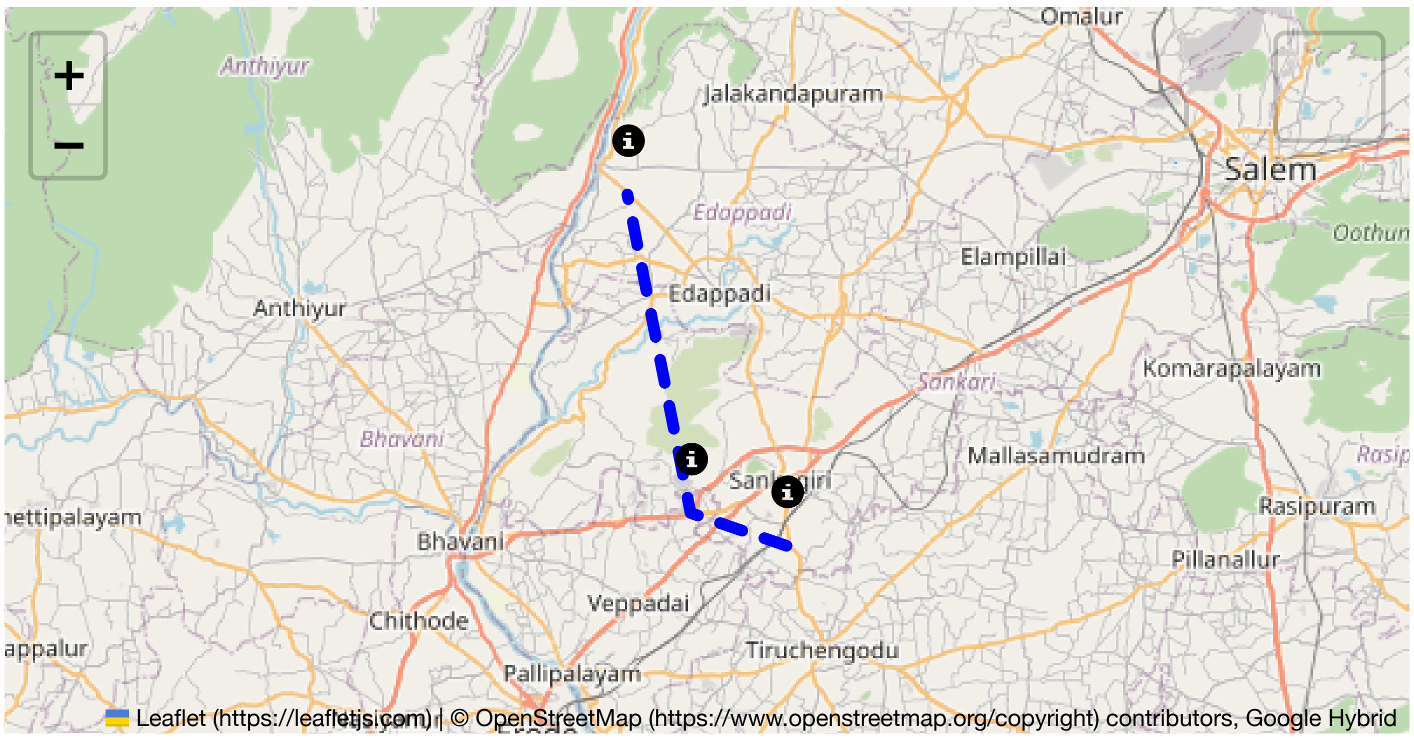
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

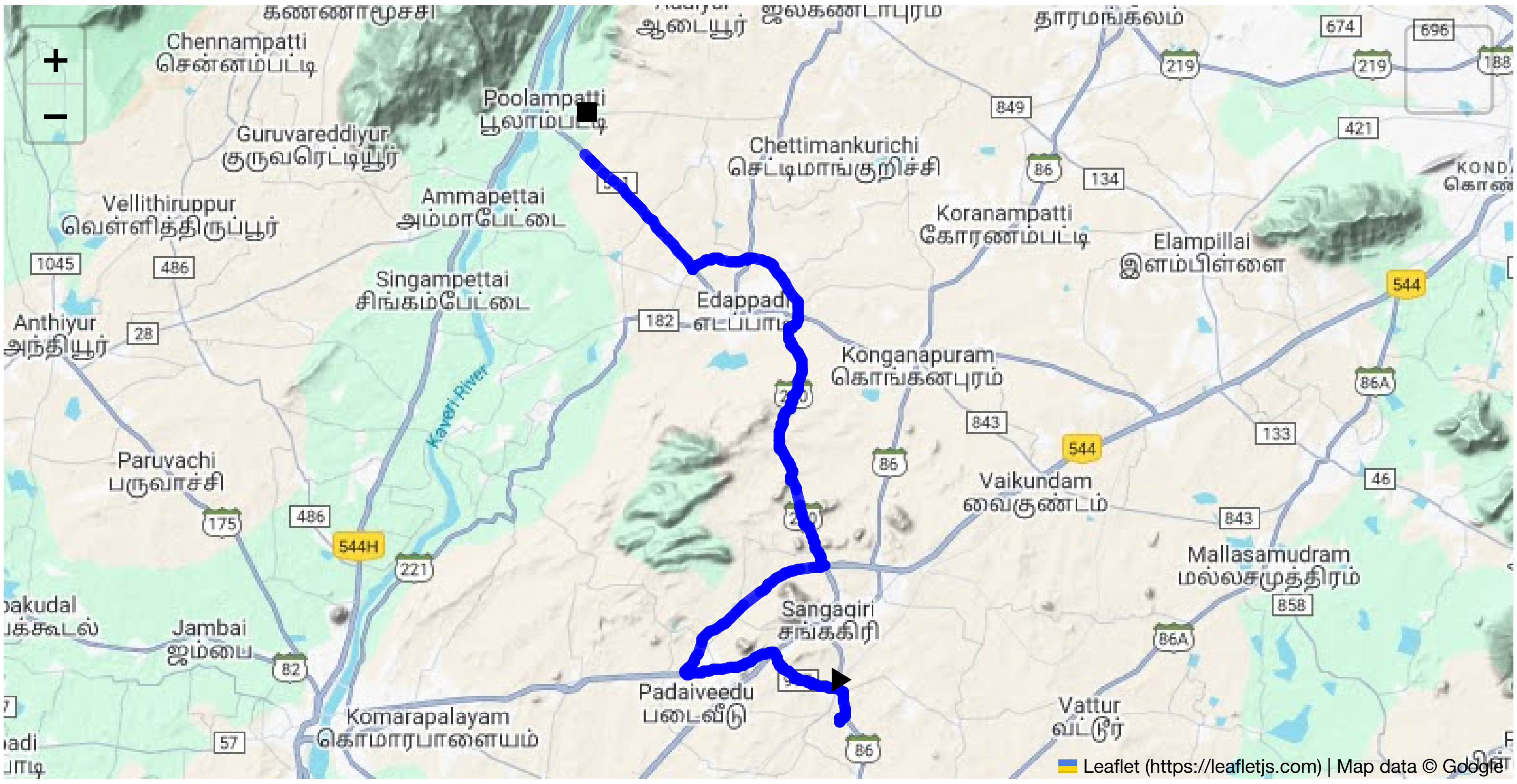
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 38.84 km
Estimated Duration: 0.9 hours
Adjusted Duration (Heavy Vehicle): 1.1 hours

Start: (11.4381, 77.8734)
End: (11.6408, 77.7806)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Poolampatti via Bhavani Main Road covers approximately 38.84 kilometers. This stretch of road is commonly used by local traffic and some commercial vehicles, including transporters of hazardous materials. The journey typically takes around 52 minutes under normal conditions for heavy vehicles.

Typical Weather Conditions

- **Weather**: The region experiences a tropical climate. Summers can be extremely hot, with temperatures reaching above 40°C, while winters are mild.
- **Rainfall**: The monsoon season from June to September can bring heavy rains, which may cause flooding and slippery road conditions.
- **Hazards**: Visibility can be significantly reduced due to rain or fog during certain times of the year.

Traffic Patterns

- **Peak Hours**: Congestion is more likely during morning hours (8-10 AM) and evening (5-7 PM) when commuter traffic is highest.
- **Congestion Areas**: The towns along Bhavani Main Road, particularly around marketplaces, can be choke points.

Road Quality and Infrastructure

- **Road Conditions**: Generally, the roads are in fair condition, but sections may have potholes or uneven surfaces particularly after the monsoon season.

- **Infrastructure**: Limited facilities for heavy vehicles, with few designated stops or repair services along the route.

Alternative Routes

- **Option 1**: If Bhavani Main Road is blocked, you might consider taking the State Highway 86, which runs parallel but might add considerable time.
- **Option 2**: Depending on the incident location, rural backroads can be used for detours, although these are not recommended for heavy vehicles.

Local Regulations on Hazardous Transport

- **Permits**: Must comply with Tamil Nadu state regulations on hazardous material which require specific permissions and adherence to safety standards.
- **Restrictions**: Transport through certain narrow or densely populated areas might be restricted or require special permits.

Historical Incidents

- No significant incidents directly on this route have been reported, but areas around Bhavani Road have seen occasional truck overturns due to overloading or speeding.

Environmental Considerations

- **Sensitive Areas**: Nearby agricultural lands imply special care to avoid spillage of hazardous materials.
- **Wildlife Crossings**: Be aware of potential wildlife crossings particularly in less populated stretches.

Communication Coverage

- **Coverage**: Major telecom providers offer coverage along most of the route; however, there are potential dead zones, especially in rural stretches.
- **Dead Zones**: Limited coverage expected near Padaiveedu.

Emergency Response

- **Response Times**: Emergency services are available in the towns along the route, with estimated response times ranging from 15 to 30 minutes, depending on proximity.
- **Facilities**: The towns have basic medical and towing services.

Overall Risk Assessment

The assessment identifies the route as moderately safe but recommends caution due to potential weather-related hazards, road quality issues, and traffic congestion points. It's crucial for drivers to stay vigilant, especially during adverse weather and peak hours. Familiarity with alternate routes and communication with local authorities for real-time updates is advised for the timely identification of potential obstructions or emergencies.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60019, 77.81968	15 KM/Hr
2	Turn	High	11.43811, 77.87348	15 KM/Hr
3	Turn	Medium	11.43956, 77.87340	30 KM/Hr
4	Turn	High	11.43968, 77.87345	15 KM/Hr
5	Turn	High	11.44029, 77.87544	15 KM/Hr
6	Turn	Medium	11.45353, 77.85697	30 KM/Hr
7	Turn	Medium	11.45839, 77.85143	30 KM/Hr
8	Turn	Medium	11.46307, 77.84941	30 KM/Hr
9	Turn	Medium	11.46316, 77.84921	30 KM/Hr
10	Turn	High	11.45542, 77.81725	15 KM/Hr
11	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
12	Blind Spot	Blind Spot	11.49390, 77.86748	10 KM/Hr
13	Turn	Medium	11.55460, 77.85624	30 KM/Hr
14	Turn	High	11.57903, 77.85432	15 KM/Hr
15	Turn	Medium	11.58022, 77.85744	30 KM/Hr
16	Turn	Medium	11.60348, 77.83967	30 KM/Hr
17	Turn	Medium	11.60345, 77.83963	30 KM/Hr
18	Turn	Medium	11.60345, 77.83958	30 KM/Hr
19	Turn	Medium	11.60347, 77.83954	30 KM/Hr
20	Turn	High	11.60349, 77.83953	15 KM/Hr

Emergency Locations

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58300, 77.85813

Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.60019, 77.81968

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45353, 77.85697

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45839, 77.85143

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46307, 77.84941

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46316, 77.84921

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.45631, 77.81668

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49390, 77.86748

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55460, 77.85624

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57903, 77.85432

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58022, 77.85744

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60348, 77.83967

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60345, 77.83963

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60345, 77.83958

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60347, 77.83954

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.60349, 77.83953